A BOND-DIMENSION-TWO MATRIX-PRODUCT STATE THAT IS AN EXACT ZERO-ENERGY EIGENSTATE AT EVERY LENGTH OF A NEAREST-NEIGHBOUR SPIN-$\frac{1}{2}$ CHAIN

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Abstract. We exhibit a translation-invariant matrix-product state (MPS) of bond dimension two, with integer transfer matrices $A^0 = \left[\begin{smallmatrix}1 & 0\\ 0 & -1\end{smallmatrix}\right]$ and $A^1 = \left[\begin{smallmatrix}-2 & -1\\ 1 & 0\end{smallmatrix}\right]$, that is an exact zero-energy eigenstate, at *every* finite length L, of the periodic nearest-neighbour spin-$\frac{1}{2}$ Hamiltonian $H = -\sum_i (I + X_i)(X_{i+1} + Z_{i+1})$. The proof is a single finite identity: a telescoping certificate $\sum_{s,t} \langle uv|h|st\rangle A^s A^t = C^u A^v - A^u C^v$ holds with the integer matrices $C^0 = \left[\begin{smallmatrix}-2 & 1\\ 1 & -2\end{smallmatrix}\right]$, $C^1 = \left[\begin{smallmatrix}3 & 2\\ 0 & -1\end{smallmatrix}\right]$ (sixteen scalar equations, all exact), whose right-hand side telescopes around the periodic trace, giving $H|\psi_L\rangle = 0$ for all L at once. This is the matrix-product-ansatz mechanism of Derrida, Evans, Hakim and Pasquier (1993) in its eigenstate form (Gehrmann and Essler, arXiv:2605.03020, Eq. (10)); the existence of a certificate of this form for any injective MPS eigenstate is the local characterization of Garre Rubio, Molnár, Schuch and Verstraete (arXiv:2603.28349), so the *technique* is not new and is credited as such. The state is genuinely bond dimension two ($\{I, A^0, A^1, A^0A^1\}$ spans $M_2(\mathbb{C})$): its exact odd/even sublattice Schmidt rank over $\mathbb{Q}$ is $2, 4, 8$ at $L = 4, 6, 8$, so the minimal number of product states in a decomposition across that cut grows over the computed range. The certificate, the eigenvalue property, the Schmidt ranks and the spanning property are re-established from raw data by a standard-library exact-arithmetic verifier; the eigenvalue property and the norms $\||\psi_L\rangle\|^2 = 4^L$ are recomputed for $L = 3, \ldots, 9$ by an independent dense `numpy` build; both codes are shipped with SHA-256 digests. We make no claim about the thermal or non-thermal character of the remainder of the spectrum of H, and object-level novelty rests on a finite sweep and cannot be absolute (§1.4).

1. Introduction

1.1. **The construction.** Fix the single-site basis $|0\rangle, |1\rangle$ and the Pauli matrices $X = \left[\begin{smallmatrix}0 & 1\\ 1 & 0\end{smallmatrix}\right]$, $Z = \left[\begin{smallmatrix}1 & 0\\ 0 & -1\end{smallmatrix}\right]$, $I = \left[\begin{smallmatrix}1 & 0\\ 0 & 1\end{smallmatrix}\right]$. For a chain of L sites with periodic boundary conditions (site $L + 1$

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Computation and authorship. The construction, the certificate, and the verifiers in this note were produced by **Claude Fable 5** (Anthropic), directed by the author, on a single Apple M4 laptop. The exact-arithmetic verifier `reverify.py` imports only the Python standard library; the independent cross-check `xcheck.py` assembles the Hamiltonian densely from Pauli matrices in `numpy` with `object`-dtype (exact integer) arithmetic and shares no code with `reverify.py`. This is a factual methods statement: the artifacts are designed so that the provenance of the *search* is irrelevant to the validity of the *result*, which is a finite exact identity together with an elementary algebraic argument.

Prior-art record. The four primary sources cited for the certificate technique and the model class — arXiv:2503.16327, arXiv:2605.03020, arXiv:2603.28349, and arXiv:1910.06616 (PRX **10**, 021051, 2020) — were read directly on August 6, 2026. The telescoping matrix-product-ansatz certificate is a published, standard technique and is credited to it at every point of use; only the specific Hamiltonian–eigenstate pair is offered as new, and only to the extent that a finite literature sweep of that date can support (§1.4). *Draft of August 6, 2026.*

identified with site 1), let

$$H \;=\; -\sum_{i=1}^{L}(I+X_i)\,(X_{i+1}+Z_{i+1}), \tag{1}$$

where $(I+X_i)(X_{i+1}+Z_{i+1})$ denotes the two-site operator $(I+X)\otimes(X+Z)$ acting on the neighbouring pair $(i,i+1)$ and the identity elsewhere. Both factors are real symmetric, so H is Hermitian.

Define the two integer 2×2 transfer matrices

$$A^0 = \begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}, \qquad A^1 = \begin{bmatrix}-2 & -1\\ 1 & 0\end{bmatrix}, \tag{2}$$

and the translation-invariant, periodic matrix-product state

$$|\psi_L\rangle \;=\; \sum_{s\in\{0,1\}^L} \mathrm{Tr}\big(A^{s_1}A^{s_2}\cdots A^{s_L}\big)\,|s_1 s_2\cdots s_L\rangle. \tag{3}$$

The bond dimension is 2: the auxiliary index summed in the trace ranges over two values. Note $A^0 = Z$; the second matrix A^1 is not normal and $\{A^0, A^1\}$ do not commute.

1.2. **Result.**

Theorem 1.1. *For every $L\ge 3$ the state $|\psi_L\rangle$ of (3) is nonzero and satisfies $H|\psi_L\rangle = 0$ exactly, with H the periodic Hamiltonian (1). The state is genuinely of bond dimension two $\big($the four matrices I, A^0, A^1, A^0A^1 are linearly independent and span $M_2(\mathbb{C})\big)$, and its exact Schmidt rank across the odd/even sublattice bipartition is $2, 4, 8$ at $L = 4, 6, 8$ respectively, so the minimal number of product states in a decomposition across that cut is at least eight at $L = 8$ and grows over the computed range.*

The eigenvalue statement $H|\psi_L\rangle = 0$ for all L is not a numerical observation extrapolated from small sizes: it follows from a single finite identity in the 2×2 matrices, proved in §2 (Lemma 2.1 and Proposition 2.2). Sizes $L = 3,\dots,10$ are additionally checked by direct dense substitution as corroboration (§4). The norm is $\||\psi_L\rangle\|^2 = 4^L$ for $L = 3,\dots,9$ (§4), so $|\psi_L\rangle \neq 0$ there; nonvanishing for all L also follows from bond-dimension-two irreducibility (Proposition 2.3).

1.3. **Provenance of the method, and what is new.** The mechanism behind Theorem 1.1 is the *matrix-product ansatz*. In the exact solution of the asymmetric simple exclusion process, Derrida, Evans, Hakim and Pasquier [DEHP93] represent the stationary weight as a matrix product and reduce stationarity to a local algebraic relation between the representing matrices; the same device produces *eigenstates*, not only steady states, when the local relation is read as a commutator identity. Gehrmann and Essler [GE26] write this eigenstate form explicitly — their Eq. (10), $h\,A\,A = E\,A - A\,E$, a generalization of the DEHP ansatz — and use it to construct exact MPS eigenstates of several spin-S chains and square-lattice models. Garre Rubio, Molnár, Schuch and Verstraete [GRMSV26] prove that a local, fixed-size equation of exactly this shape — how one Hamiltonian term acts on a block of tensors — is *necessary and sufficient* for an injective MPS to be an exact eigenstate of an extensive local operator.

Two consequences frame this note precisely. First, the *certificate technique is not new*: it is standard, and the existence of a certificate of the form used here is guaranteed *a priori* for any injective MPS eigenstate by [GRMSV26]. We credit it as such at the point of use in §2, and claim no novelty in the method. Second, what a certificate does not do is exhibit the pair; the content offered here is the specific Hamiltonian–eigenstate pair (1)–(3) together with its explicit integer certificate (5) and the entanglement facts of §3. The scope and the limits of that object-level claim are stated in §1.4.

1.4. **What is not claimed.** The claim is object-level and is bounded in three explicit ways.

The technique is credited, not claimed. As above, the telescoping matrix-product-ansatz certificate is due to [DEHP93] in origin, is written in eigenstate form in [GE26], and is guaranteed to exist by [GRMSV26]. Nothing about the *method* is asserted to be new.

No spectral or thermalization claim is made. We establish that $|\psi_L\rangle$ is one exact eigenstate, with eigenvalue 0, of H. We do not analyse the rest of the spectrum of H, its level-spacing statistics, or whether the model thermalizes; no statement about non-thermal behaviour, weak ergodicity breaking, or the surrounding spectrum is made or implied. The proven content is exactly the four properties of Theorem 1.1.

Object-level novelty rests on a finite sweep. On August 6, 2026 the four sources above were read directly. In [IM25] the exact area-law eigenstates are constructed inside a kinetically constrained subspace (PXP-type models, via a projector onto a Fibonacci-constrained space); our $|\psi_L\rangle$ lives in the full 2^L-dimensional Hilbert space with no projector-defined subspace, and its transfer matrices (2) are not among theirs. The spin-$\frac{1}{2}$ instance of Model I of [GE26] is a Rydberg-plus-Dzyaloshinskii–Moriya chain with a complex free-parameter family of tensors, a different Hamiltonian with non-integer tensors; the worked example of [GRMSV26] is the quantum-group symmetry of the XXZ chain; and the states of [PGCGB20] become exact only in the large-size limit, whereas $|\psi_L\rangle$ is exact at every finite L. In none of the four was the pair (1)–(3) found. A finite sweep cannot exclude a unitarily or gauge-equivalent reformulation in a venue not examined; the novelty claim is made at exactly that strength and no more.

2. The telescoping certificate and the all-length proof

2.1. **The bond operator.** Write the two-site bond operator of (1) as

$$h \;=\; -(I+X)\otimes(X+Z), \tag{4}$$

so that $H=\sum_{i=1}^{L} h_{i,i+1}$ (periodic). In the ordered two-site basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$,

$$h \;=\; -\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \end{bmatrix},$$

which is real symmetric, confirming $H^\dagger = H$. We index its entries by $\langle uv|h|st\rangle$ with $u,v,s,t \in \{0,1\}$.

2.2. **The certificate identity.** The following is the matrix-product-ansatz certificate in the eigenstate form of [GE26] (Eq. (10)), whose existence for an injective MPS eigenstate is guaranteed by [GRMSV26] and whose mechanism is that of [DEHP93].

Lemma 2.1 (Certificate)**.** *With h as in (4), A^0, A^1 as in (2), and the integer matrices*

$$C^0 = \begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix}, \qquad C^1 = \begin{bmatrix} 3 & 2 \\ 0 & -1 \end{bmatrix}, \tag{5}$$

the sixteen scalar equations

$$\sum_{s,t\in\{0,1\}} \langle uv|h|st\rangle\, A^s A^t \;=\; C^u A^v - A^u C^v \qquad (u,v\in\{0,1\}) \tag{6}$$

hold exactly.

Proof. Both sides of (6) are 2×2 integer matrices, so this is a finite verification. For $(u, v) = (0, 0)$ the row $\langle 00|h|\cdot\rangle$ is $-(1, 1, 1, 1)$, so the left-hand side is $-(A^0A^0 + A^0A^1 + A^1A^0 + A^1A^1) = -\left[\begin{smallmatrix} 0 & 2 \\ -2 & 0 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 0 & -2 \\ 2 & 0 \end{smallmatrix}\right]$, while $C^0A^0 - A^0C^0 = \left[\begin{smallmatrix} -2 & -1 \\ 1 & 2 \end{smallmatrix}\right] - \left[\begin{smallmatrix} -2 & 1 \\ -1 & 2 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 0 & -2 \\ 2 & 0 \end{smallmatrix}\right]$, which agree. The remaining fifteen equations are identical finite checks. The verifier `reverify.py` (§4) does not assume (5): it solves the linear system (6) for the eight entries of C^0, C^1 exactly over $\mathbb{Q}$, finds it consistent, recovers (5), and confirms all sixteen residuals vanish. $\square$

2.3. **The periodic telescope.**

Proposition 2.2. *For every $L \geq 1$, $H|\psi_L\rangle = 0$.*

Proof. Fix a basis configuration $u = (u_1, \ldots, u_L) \in \{0, 1\}^L$. The amplitude of $|\psi_L\rangle$ at u is $\mathrm{Tr}(A^{u_1} \cdots A^{u_L})$. Applying the single bond term $h_{i,i+1}$ and reading off the amplitude at u replaces the adjacent pair $A^{u_i}A^{u_{i+1}}$ inside the trace by $\sum_{s,t}\langle u_iu_{i+1}|h|st\rangle A^sA^t$; by the certificate (6) this equals $C^{u_i}A^{u_{i+1}} - A^{u_i}C^{u_{i+1}}$. Hence

$$(h_{i,i+1}|\psi_L\rangle)_u = \mathrm{Tr}\big(A^{u_1} \cdots A^{u_{i-1}}\left[C^{u_i}A^{u_{i+1}} - A^{u_i}C^{u_{i+1}}\right]A^{u_{i+2}} \cdots A^{u_L}\big) = f(i) - f(i+1),$$

where $f(i)$ denotes the trace $\mathrm{Tr}(A^{u_1} \cdots A^{u_{i-1}}\, C^{u_i}\, A^{u_{i+1}} \cdots A^{u_L})$ with the single insertion $A^{u_i} \mapsto C^{u_i}$ at position i, and where the second term is $f(i+1)$ because a C at the right factor of the bond $(i, i+1)$ is a C at position $i + 1$. Summing over the periodic chain,

$$(H|\psi_L\rangle)_u = \sum_{i=1}^{L}\big(f(i) - f(i+1)\big) = f(1) - f(L+1) = 0,$$

the sum telescoping and $f(L + 1) = f(1)$ by periodicity of the trace. As u was arbitrary, $H|\psi_L\rangle = 0$. $\square$

The argument uses only the finite identity (6) and the cyclic invariance of the trace; it is uniform in L. This is precisely the telescoping guaranteed by [GRMSV26] for an MPS eigenstate and realized concretely as in [GE26, DEHP93].

2.4. **Genuine bond dimension two, and non-frustration-freeness.**

Proposition 2.3. *The matrices I, A^0, A^1, A^0A^1 are linearly independent and span $M_2(\mathbb{C})$; equivalently, the MPS (3) is injective of bond dimension exactly two. Consequently $|\psi_L\rangle \neq 0$ for all $L \geq 3$.*

Proof. Writing each 2×2 matrix as a row vector in the basis $(E_{11}, E_{12}, E_{21}, E_{22})$ gives $I = (1, 0, 0, 1)$, $A^0 = (1, 0, 0, -1)$, $A^1 = (-2, -1, 1, 0)$ and $A^0A^1 = (-2, -1, -1, 0)$; the determinant of the resulting 4×4 integer matrix is $-4 \neq 0$, so the four matrices are a basis of $M_2(\mathbb{C})$. Spanning $M_2(\mathbb{C})$ is injectivity of the MPS transfer map, which forces $|\psi_L\rangle \neq 0$ for $L \geq 3$; the explicit norms $\||\psi_L\rangle\|^2 = 4^L$ of §4 confirm this for $L = 3, \ldots, 9$. $\square$

Remark 2.4 (Not frustration-free). The cancellation in Proposition 2.2 is global, not termwise: a single bond term does not annihilate the state. Explicitly $h_{1,2}|\psi_4\rangle \neq 0$ (checked exactly; §4), so $|\psi_L\rangle$ is not a common zero of the individual bond terms and H is not frustration-free on it. The eigenvalue 0 is produced by the telescoping sum of nonzero local contributions, which is the whole point of the certificate.

3. Entanglement: a growing sublattice Schmidt rank

For a bipartition of the sites into two blocks, the Schmidt rank of a state is the rank of its coefficient matrix under the induced reshaping; a state that is a sum of k product states has Schmidt rank at most k across *every* bipartition. Two bipartitions are computed exactly over $\mathbb{Q}$ (§4).

Proposition 3.1. *For the contiguous half-cut* $\{1,\dots,L/2\} \mid \{L/2+1,\dots,L\}$*, the exact Schmidt rank of* $|\psi_L\rangle$ *is* 4 *at* $L = 4, 6, 8$*, saturating the bond-dimension bound* $D^2 = 4$*. For the odd/even sublattice bipartition* $\{1,3,5,\dots\} \mid \{2,4,6,\dots\}$*, the exact Schmidt rank of* $|\psi_L\rangle$ *is* $2, 4, 8$ *at* $L = 4, 6, 8$ *respectively.*

The contiguous half-cut rank equalling $D^2 = 4$ is the generic MPS value and is consistent with an area law along a contiguous cut. The odd/even rank is the substantive fact: it *increases* with L over the computed range, $2 \to 4 \to 8$. Since a sum of k product states has Schmidt rank at most k across every bipartition, the odd/even Schmidt rank equals the minimal number of product states in any decomposition across that cut; thus $|\psi_8\rangle$ requires at least eight such product states, and this minimal number grows over $L = 4, 6, 8$. The computed values follow the pattern $2^{L/2-1}$; we state the three exact values as proven and record the growth as observed at $L = 4, 6, 8$, not established for all L, and we make no L-uniform product-state-complexity claim for the family $\{|\psi_L\rangle\}$.

Remark 3.2 (A one-sided rotated form). The global single-site rotation U with $UXU^{-1} = Z$, $UZU^{-1} = -X$ sends $I + X \mapsto I + Z = 2P_0$, with $P_0 = |0\rangle\langle 0| = (I+Z)/2$, and $X + Z \mapsto Z - X$, so

$$UHU^{-1} \;=\; -2\sum_{i=1}^{L} P_0^{(i)}\,(Z - X)_{i+1},$$

a one-sided facilitated ("East-type") form in which site $i+1$ is acted on conditioned on site i (verified by exact conjugation; §4). This is East-type only in the structural sense of a one-sided facilitation; it is not the quantum East model of Pancotti, Giudice, Cirac, Garrahan and Bañuls [PGCGB20] (whose non-thermal states become exact eigenstates only in the large-size limit, in contrast to the exact-at-every-L state here), and it carries an additional density-density term relative to the standard East interaction. We record the reformulation as a structural observation and draw no dynamical conclusion from it.

4. Verification and reproduction

The quantitative claims of Theorem 1.1 and §3 are re-established from the raw data (2)–(5) by two independent codes, shipped with the note and pinned by SHA-256 in Table 1; the coverage of each is itemized below. The one exception is the determinant -4 of Proposition 2.3, which neither code computes. The verifier establishes rank four, which is equivalent to the determinant being nonzero — what the proposition's argument requires — but not to its value. The certificate's `genuine_bond_dimension_2` field attributes the value -4 to `reverify.py` P5; P5 verifies the equivalent rank-4 condition. The certificate is left byte-identical to its hash-pinned shipped form; the attribution is corrected here.

Exact-arithmetic verifier (`reverify.py`, Python standard library only; no third-party imports; no POSIX-only calls). It (i) forms h from I, X, Z and checks it real symmetric; (ii) *solves* the linear system (6) for C^0, C^1 exactly over $\mathbb{Q}$, finds it consistent, recovers (5), and confirms all sixteen residuals vanish (Lemma 2.1); (iii) builds $|\psi_L\rangle$ from (3) and applies H densely in exact integers, obtaining $H|\psi_L\rangle = 0$ and $|\psi_L\rangle \neq 0$ for $L = 3,\dots,10$; (iv) computes the exact Schmidt ranks of Proposition 3.1 over $\mathbb{Q}$; (v) confirms I, A^0, A^1, A^0A^1 span $M_2(\mathbb{C})$ (Proposition 2.3); (vi) confirms $h_{1,2}|\psi_4\rangle \neq 0$ (Remark 2.4); and (vii) confirms the rotation identity of Remark 3.2 by exact integer conjugation. It reports `ALL CHECKS PASS` (20/20).

Independent cross-check (`xcheck.py`). A from-scratch `numpy` build sharing no code with `reverify.py`: the dense Hamiltonian is assembled directly from Pauli matrices by Kronecker products, the amplitudes from explicit $\mathrm{Tr}(A^{s_1} \cdots A^{s_L})$, all in `object`-dtype (arbitrary-precision integer) arithmetic. For $L = 3, \ldots, 9$ it prints $H|\psi_L\rangle = 0$ exactly, $|\psi_L\rangle \neq 0$, H real symmetric, and the exact squared norm, which is 4^L at each of those lengths. The script reports these values without asserting them — it contains no failure path and exits 0 unconditionally — so the comparison with 4^L is made against its printed output. Two disjoint code paths agree.

The trusted base. The all-length statement (Theorem 1.1, eigenvalue part) rests on the finite identity (6) — sixteen integer equations — and the elementary telescoping of Proposition 2.2; there is no numerical extrapolation in it and no solver in its trusted base. The dense checks at $L = 3, \ldots, 10$ are corroboration, not the proof. What must be trusted is minimal: (i) exact-integer/rational arithmetic (CPython for `reverify.py`; `numpy` `object`-dtype for `xcheck.py`); (ii) the shared convention, stated in `object.json`, for reading the amplitude (3) and the operator ordering of (4) — a consistent misconvention on both sides would not be caught by re-checking, but the two codes fix the operator independently from Pauli matrices; (iii) the object-level novelty of §1.4, which is a finite literature sweep and cannot be absolute. The searcher's provenance is *not* in the trusted base: the certificate is re-derived by an independent exact linear solve and the eigenvalue is re-checked by two disjoint dense builds.

To reproduce on any machine with Python 3 (and, for the cross-check, `numpy`):

```
python3 reverify.py     # stdlib only; prints ALL CHECKS PASS (20/20)
python3 xcheck.py       # independent numpy build; H@psi==0, ||psi||^2=4^L
shasum -a 256 reverify.py xcheck.py object.json   # match Table 1
```

On a platform whose hashing tool is `sha256sum` rather than `shasum`, substitute `sha256sum reverify.py xcheck.py object.json`.

file	SHA-256
`reverify.py`	`e57a0aea13d348311fcd2e260474fb40b68a80c60ed12840a5c03718ff9179d9`
`xcheck.py`	`f43eb90b738f74fef71650fd1b7c78f9a7ff706f31e367fefa3843a6bbef4fc2`
`object.json`	`1353baa37f960332ad3b6a8013d57c92168befd18be3323b4f16309ae13076a3`

Table 1. SHA-256 digests of the shipped verifier, cross-check, and machine-readable object specification. `object.json` records (2), (4), (5), the amplitude and operator conventions, and the verified properties.

Acknowledgments

The matrix-product ansatz is Derrida, Evans, Hakim and Pasquier's; its eigenstate form used here is written explicitly by Gehrmann and Essler, and the local necessary-and-sufficient characterization that guarantees a certificate of this shape for any injective MPS eigenstate is Garre Rubio, Molnár, Schuch and Verstraete's. The quantum East model referenced in Remark 3.2 is Pancotti, Giudice, Cirac, Garrahan and Bañuls's. The computation and drafting were AI-assisted as stated in the first footnote.

References

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[GRMSV26] J. Garre Rubio, A. Molnár, N. Schuch and F. Verstraete, *The local characterization of global tensor network eigenstates*, arXiv:2603.28349 (2026); read directly on August 6, 2026. A local, fixed-size equation — how one operator term acts on a block of tensors — is proved necessary and sufficient for an injective MPS to be an exact eigenstate of an extensive local operator; the worked example is the quantum-group symmetry of the XXZ chain.

[IM25] A. A. Ivanov and O. I. Motrunich, *Many exact area-law scar eigenstates in the nonintegrable PXP and related models*, arXiv:2503.16327 (2025); read directly on August 6, 2026. Exact area-law eigenstates are constructed inside a kinetically constrained subspace (PXP-type models), via a projector onto a Fibonacci-constrained space; the present $|\psi_L\rangle$ lives in the full 2^L Hilbert space with no such projector.

[PGCGB20] N. Pancotti, G. Giudice, J. I. Cirac, J. P. Garrahan and M. C. Bañuls, *Quantum East model: localization, non-thermal eigenstates and slow dynamics*, Phys. Rev. X **10** (2020), 021051; also arXiv:1910.06616. Read directly on August 6, 2026. Its non-thermal states become exact eigenstates in the large-size limit, in contrast to the exact-at-every-finite-L state of Theorem 1.1.

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